Fig 5c Cross-axis convergence classification (Path A strict)

v3 honest take: AF molecular endotypes are fragmentary at the family level; only 1 family reaches triple-axis convergence (2 double, 7 single, 6 silent).

Path A definition: L1 supportive = replication_class = high (CI excludes 0); L2 supportive = fidelity_class ≠ Below_detection; L3 supportive = chronicity_class ≠ Stable.

a Cross-axis convergence (n = 16 families, Path A strict)

b Per-family evidence stack

Family		Cross-axis class	L1	L2	L3	n.sup	Subtype
F15	NTRK2 / TrkB signalli…	Triple-positive	+	+	+	3/3	L1+L2+L3
F11	BMP signalling	Single-axis	+			1/3	L1
F9	HS-GAG metabolism	Double-positive	+		+	2/3	L1+L3
F10	TGF-beta signalling (H	Double-positive	+		+	2/3	L1+L3
F16	Keratan sulfate biosy…	Single-axis		+		1/3	L2
F3	TG metabolism	Single-axis			+	1/3	L3
F14	Eukaryotic translatio…	Silent				0/3	none
F2	Mito Ca2+ transport	Single-axis			+	1/3	L3
F12	rRNA processing	Silent				0/3	none
F1	Ribosome QC (RQC)	Silent				0/3	none
F5	tRNA aminoacylation	Silent				0/3	none
F4	BCAA catabolism	Silent				0/3	none
F13	Amino-acid metabolism	Single-axis			+	1/3	L3
F7	TCA cycle	Single-axis			+	1/3	L3
F8	Complex I biogenesis	Single-axis			+	1/3	L3
F6	Mito protein import	Silent				0/3	none

c Members per cross-axis class (with supportive-axis subtype)

Triple-positive (1) — *all three axes supportive*

F15 NTRK2 / TrkB signall [L1+L2+L3]

Double-positive (2) — *two of three axes supportive*

F 9 HS-GAG metabolism [L1+L3]
F10 TGF-beta signalling [L1+L3]

Single-axis (7) — *exactly one axis supportive*

F11 BMP signalling [L1]
F16 Keratan sulfate bios [L2]
F 3 TG metabolism [L3]
F 2 Mito Ca2+ transport [L3]
F13 Amino-acid metabolis [L3]
F 7 TCA cycle [L3]
F 8 Complex I biogenesis [L3]

Silent (6) — *no axis supportive (strict null; absence of evidence ≠ evidence of absence)*

F14 Eukaryotic translati [none]
F12 rRNA processing [none]
F 1 Ribosome QC (RQC) [none]
F 5 tRNA aminoacylation [none]
F 4 BCAA catabolism [none]
F 6 Mito protein import [none]